

MATH 565 Monte Carlo Methods

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Final Exam—Take Home Part

Due by 2 PM, December 11, 2025

Instructions:

- i. This is a **take-home** exam with **ONE** question(s). Attempt all. The maximum number of points is **35**.
- ii. This take-home exam is due at the time listed above. There is no time limit other than the due date.
- iii. You must turn in your files via Canvas and hard copies to the instructor.
- iv. You may use AI tools (e.g., ChatGPT) and other inanimate resources. You may **not** get help from any human (classmates, tutors, friends, family, or online responders). All solutions must be your own work, and you are responsible for checking that AI-generated content is correct.
- v. Show all your work to justify your answers. Answers without adequate justification will not receive credit.
- vi. If you have clarification questions, email the instructor at hickernell@illinoistech.edu. Mathematical help will not be given.

I understand these instructions and have not relied on any help for this exam beyond what is allowed, nor provided any help to anyone else beyond what is allowed.

Signature

Date

1. (35 points) This problem concerns modeling the price of a financial option. We model the price of a stock in terms of a geometric Brownian motion, as

$$S(t_j, X_j) = S(0) \exp((r - \sigma^2/2)t_j + \sigma X_j),$$
$$t_j = jT/d, \quad (X_1, \dots, X_d)^\top \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad \Sigma = (\min(t_j, t_k))_{j,k=1}^d$$

We consider two kinds of option *payoffs* with the discount factor:

Asian arithmetic mean call: $\text{payoff}(\mathbf{X})$

$$= \max\left(\frac{1}{d}[S(0)/2 + S(t_1, X_1) + \dots + S(t_{d-1}, X_{d-1}) + S(T, X_d)/2] - K, 0\right) \times \exp(-rT)$$

$$\text{Lookback call: } \text{payoff}(\mathbf{X}) = \max\left(S(T, X_d) - \min_{0 \leq j \leq d} (S(t_j, X_j))\right) \times \exp(-rT)$$

The basic parameters that we will use are

time to maturity $T = 1/4$ (quarter of a year)
number of time steps $d = 13$ (weekly steps)
initial stock price $S(0) = 50$
interest rate $r = 0.02$
volatility $\sigma = 0.40$
strike price $K = 50$ (at the money)

The option price is the expected value of the discounted payoff:

$$\text{price} = \mathbb{E}[\text{payoff}(\mathbf{X})].$$

- a. (20 points) What prices do you get for the two different options using $n = 2^{17}$ IID samples, and what are your confidence intervals for the prices?
- b. (8 points) What prices do you get for the two different options using $n = 2^{17}$ Sobol' samples, and what are your confidence interval for the prices? How do these confidence intervals compare to those for IID sampling?
- c. (7 points) If you change your monitoring to daily ($d = 91$), how do the option prices change in part a., and why? How does the time required change? Explain these changes in prices and runtime.